

AI-Mediated DIY Memorialization: Exploring AI-Generated Videos For Remembrance and Continuing Bonds

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1 Introduction

Recent advancements in generative AI have enabled the synthesis of hyper-realistic images, videos, audio, and multi-modal content. Such AI-generated media create new opportunities for interacting with digital portrayals of individuals, from fictional characters to historical figures [26], and even deceased family members [36]. These generative techniques broaden how people imagine representing or reconnecting with others through digital media.

Maintaining healthy continuing bonds with the deceased has long been recognized as beneficial for the bereaved [14]. Over the past decades, researchers in the Human-Computer Interaction (HCI) community have explored how to maintain such bonds through tangible and digital items left by the deceased, which were broadly defined as legacy [6], and through technologies designed to support remembrance and bereavement in everyday contexts [13, 16].

Recent studies have begun to explore AI-generated agents that simulate the deceased, often referred to as AI Afterlife [17, 22]. These systems can generate new information, interact with others, and even evolve over time. While such agents may offer continued presence and ongoing connection [17], their generative, interactive, and evolving nature introduces substantial emotional and ethical risks for the living, such as emotional dependency and blurred boundaries between simulation and reality [17, 22].

Given the opportunities of generative AI and the risks of interactive agents, this study focuses on non-interactive AI-generated media, particularly video, as a medium for memorialization rather than an interactive recreation of the deceased. It highlights how users themselves engage in a curation process supported by generative AI, which is described as **AI-mediated Do-It-Yourself (DIY) memorialization**. This approach gives users greater authorship and agency over how memories are represented, and allows them to manage emotional engagement within a more controllable scope.

However, little is known about how individuals actually engage with AI-generated videos, or how such engagement may shape remembrance and continuing bonds with the deceased. Addressing these gaps is crucial for researchers, designers, and developers to better harness the potential of generative AI to support people's everyday DIY memorialization practices. To advance this understanding, this study examined *how we can design for and reflect through AI-mediated DIY memorialization, with a focus on creating AI-generated videos to support remembrance and continuing bonds*. Specifically, we investigated three research questions:

- **RQ1:** How do individuals negotiate emotional, realistic, and ethical boundaries when creating AI-generated memorial videos?
- **RQ2:** How do family members respond to and contribute to meaning-making around AI-generated memorial videos?

- **RQ3:** How do the material behaviours and limitations of generative AI shape DIY memorialization practices?

2 Background and Related Work

This section reviews three strands of literature that frame this study: the theory of continuing bonds, posthumous technologies in HCI, and emerging work on generative AI in posthumous contexts.

2.1 Theoretical Background: Continuing Bonds

This study draws on the concept of continuing bonds [14], which proposes that maintaining a connection with the deceased is both normal and beneficial during bereavement. The theory suggests that such bonds can provide comfort and support, helping people cope with loss and adjust to life without their loved one.

This perspective challenges traditional grief models that suggest the bereaved should detach from the deceased to move on, and has motivated extensive research on understanding and designing for memorial technologies that help the bereaved maintain ongoing connections with the deceased [1–3, 5, 20, 22, 29], for example through communal mourning in online communities [3].

2.2 Posthumous Technologies in HCI

HCI researchers have extensively examined technologies created for posthumous purposes over the past few decades, including tangible or digital legacy items left by the deceased [6, 8], as well as tools that support remembrance and bereavement practices [13, 16]. The latter spans tangible artifacts [16, 35], multimedia materials [12], and more interactive systems such as digital games [16] and chatbots [36]. Memorial practices have also been explored across diverse contexts, including home [21, 30–32], online communities [3, 5, 19, 20], and ritual or institutional settings [33, 34].

Prior work has largely focused on traditional media or interactive systems. Little is known about how generative AI can support DIY creation of non-interactive memorial videos, or how such media might enable remembrance and continuing bonds.

2.3 Generative AI in Posthumous Context

Generative AI has enabled the creation of realistic media [7, 18, 28] and interactive agents capable of simulating human appearance, voice, behavior, and memory [10, 15, 17, 22, 24]. These systems have supported well-being by enabling novel co-presence and interaction experiences [26], from learning with historical figures [25, 26], to emotion regulation and companionship [10, 15, 27].

Recent work has begun to explore generative AI for posthumous purposes, including curating digital legacy [17] and coping with grief, through AI-generated simulations of the self or the deceased

[17, 22, 36]. These agents serve as a way to sustain personal identity or continue relationships. However, continuing bonds in that way raise ethical and emotional risks, such as emotional reliance and blurred boundaries between simulation and reality [17, 22, 36]. Lei et al. point out bidirectional interaction and potential proactivity are factors of such agents that cause risks, and highlights the importance of balancing support and intrusiveness [17].

To avoid these risks, this study focuses on non-interactive AI-generated videos as a more controllable, user-authored form of memorialization that supports remembrance and continuing bonds.

Summary. These strands of research highlight gaps in existing work and establish the motivation for this study. Building on continuing bonds theory, posthumous technologies, and recent work on generative AI, this study aims to contribute to HCI and design research by examining AI-generated video as a non-interactive, user-authored medium for DIY memorialization. This approach extends prior work by shifting attention from interactive agents toward controllable, bereaved-driven memorial practices that may better support remembrance and continuing bonds.

3 Method: RtD Approach Selected and Applied

3.1 Justification of the Selected Approach

This study adopts a Designer-Researcher approach within Research through Design (RtD), specifically Autobiographical Design. This method is defined as “*design research drawn from extensive, genuine use by those creating or building the system*” [23]. It involves iterative cycles of design, use, and reflection grounded in the researcher’s lived experiences, enabling close attention to both the functional and emotional aspects of technology.

Autobiographical Design is widely used in sensitive or emotionally charged domains that require careful ethical consideration, such as bereavement [35], vulnerability [11], privacy [9], and intimate life [37]. For example, Wallace et al. used autobiographical design in the creation and long-term use of ReFind, a handheld memorial artifact, to investigate how design can support ongoing connections with the deceased [35]. Prior work also shows that autobiographical design does not occur in isolation. Rather, family members, partners, or friends often become naturally embedded in the researcher’s experience and may shape the design process [4].

This method is particularly suitable for this study due to the personal and sensitive nature of memorial practices and the researcher’s genuine need to remember her late grandfather. The approach allows the researcher and proximate family members, who are naturally part of the memorial context, to engage in reflection and use without the risks associated with formally recruiting participants who may be emotionally vulnerable. By grounding the inquiry in first-person and lived familial experience, the study minimizes ethical and emotional concerns while enabling an in-depth exploration of AI-mediated DIY memorialization.

As with many Designer-Researcher approaches, Autobiographical Design also carries methodological limitations. First, the insights generated are deeply situated and not intended to be generalizable in the same way as those from studies with formally recruited participants. Second, the researcher’s positionality shapes both the design and interpretation, which introduces subjectivity but

is appropriate for exploring emotional and relational dimensions of memorial technologies. Finally, although Autobiographical Design typically involves long-term engagement [23], the scope of this course limits the current work to early design iterations and reflective documentation of the video creation process.

Overall, Autobiographical Design is well aligned with the aims of this study, enabling a grounded, ethically responsible, and emotionally attuned inquiry into AI-mediated DIY memorialization.

3.2 Study Activities

In practice, the autobiographical design process unfolded through four major trials, each focusing on a different memorial theme. Within each trial, the researcher iteratively used AI to generate videos through multiple attempts, as shown in Figure 1. Each attempt consisted of video generation and immediate micro-reflection on realism, emotional resonance, and ethical comfort, which in turn guided subsequent adjustments. When a video felt emotionally safe to share, the researcher engaged proximate family members in discussion, inviting their emotional responses, interpretations, and any memories or reflections the video evoked, which also shaped further adjustments for the next attempt.

A trial concluded when one of two conditions was met: (1) an attempt produced a version that felt acceptable to both the researcher and family members; or (2) no acceptable version could be reached even for the researcher. Both led to structured reflection (described in subsection 3.3) and a transition to the next trial.

3.3 Reflection Framework

Guided by the three research questions, the reflective analysis in each trial was structured around four analytical lenses that together captured emotional, relational, material, and conceptual dimensions of AI-mediated DIY memorialization.

- *Realistic, emotional, and ethical fit* examined how the generated video aligned with the researcher’s memories, emotional comfort, and ethical boundaries, identifying cues that felt authentic, unsettling, or unacceptable.
- *Family interpretation and dialogue* documented the emotional responses, memory corrections, reinterpretations, and new stories that emerged when videos were shared, revealing the multi-voice dynamics of family remembrance.
- *Process insights* captured what each trial revealed about the behaviour and material qualities of the AI tools, such as what aspects felt controllable, unpredictable, biased, or resistant to manipulation. This lens helped articulate the evolving design space and technical boundary conditions.
- *Conceptual fit* assessed how the trial contributed to the emerging understanding of AI-mediated DIY memorialization, including the role of AI, the nature of DIY authorship, and how the artifact supported remembrance and continuing bonds.

These lenses served as the methodological backbone of the reflection process, guiding both within-trial sensemaking and the cross-trial analysis presented in section 4.

3.4 Process Documentation and Analysis

Across the four trials, the RtD process was documented through three complementary forms of material. First, each AI-generated

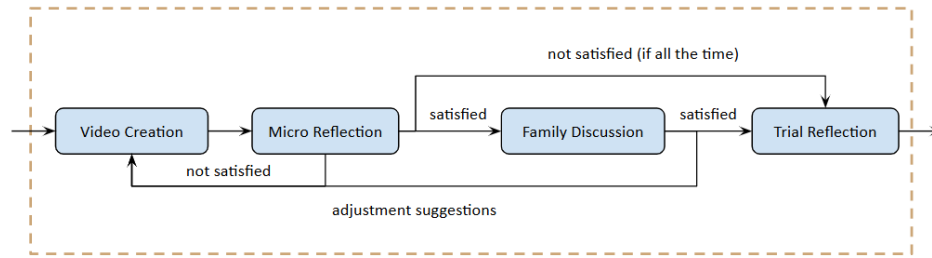


Figure 1: The iterative process within each trial: video creation, micro-reflection, conditional family discussion, and trial-level reflection.

video was archived as a design artifact representing a specific attempt consisting of specific AI tool, prompt configuration, and input materials such as photos. Second, micro-reflection notes captured the researcher’s moment-to-moment reactions, while conversational records documented family feedback and interpretations when outputs were shared. Finally, a trial-level reflective journal synthesized insights at the end of each trial using the reflection framework. These materials provide a transparent trace of design decisions, reasoning, and reflection, enabling later analysis within and across trials.

Analysis proceeded in two layers. First, observations after each trial were organized using the four-part reflection framework. Second, all trials were reviewed comparatively to identify cross-iterative patterns. Rather than thematic saturation, the goal was conceptual coherence: tracing how each trial redirected the next and how DIY memorialization practices emerged through breakdowns, adjustments, and reinterpretations.

3.5 Ethical Considerations

Given the sensitive nature of memorialization and the involvement of a deceased family member, this study required careful ethical consideration throughout the RtD process. Although the study relies on autobiographical materials and does not recruit external participants beyond family members, the visual likeness of the researcher’s late grandfather is an identifiable form of personal data. As such, the researcher chose not to include any image frames or stills from the AI-generated memorial videos in section 4.

Two reasons guided this decision. First, it is not possible to obtain informed consent from the deceased, and sharing his likeness raises questions around posthumous privacy, dignity, and representational integrity. Second, the AI-generated videos occasionally produced distortions or speculative reinterpretations of his appearance. Displaying these outputs publicly risks misrepresentation and could inadvertently violate cultural expectations surrounding respectful remembrance. Instead, all findings are reported through textual description and analysis, focusing on emotional, relational, technical, and conceptual dynamics rather than visual depiction.

4 Findings

Across four autobiographical design trials, this study generated and reflected on a series of AI-produced memorial videos that acted as design probes into AI-mediated DIY memorialization. Rather than

developing a single refined artefact, the inquiry unfolded through four distinct exploratory directions, with each experimenting with different memorial forms and revealing unique emotional, ethical, and technical dynamics of working with generative AI as a memorial material. The four trials examined:

- *Scenario Reconstruction* as a narrative memorial form;
- *Expression Animation* as a minimal memorial form;
- *Movement Rendering* as an embodied memorial form;
- *Youth Reinterpretation* as a speculative memorial form.

These explorations trace an evolving trajectory of how AI-generated videos can support remembrance and continuing bonds, and what tensions, breakdowns, and opportunities arise when a bereaved person attempts to craft DIY memorials with generative systems.

The following subsections first describe each trial as part of this iterative trajectory, and then present cross-trial insights based on the reflection framework introduced in the Method section.

4.1 Iterative Design Trajectory

Across the study, each trial unfolded through cycles of AI video generation and reflection. Micro-reflections within each trial helped identify which visual cues resonated or conflicted with lived memory, while family discussions when emotionally appropriate introduced additional interpretations and emotional layers, which both redirected the design attempts. At the end of each trial, a structured, trial-level reflection synthesized what that round of exploration had revealed and determined where the inquiry should go next, and ultimately shaped the four distinct directions of design exploration presented below.

4.1.1 Scenario Reconstruction as a Narrative Memorial Form. This initial trial examined whether a rich Lunar New Year memory could be reconstructed as a narrative memorial video. Beginning from a detailed autobiographical scene where grandfather moving through festival chores with one arm and insisting on serving soup, the researcher used Sora and later Doubao to generate multi-step videos. Prompts were gradually narrowed from full-scene descriptions to a single action (bending toward a pressure cooker), while curated photos of the grandfather, household artifacts, and contextual cues were incorporated to guide the generation.

Across attempts, the trial surfaced a strong emotional–realistic mismatch. Many outputs contained culturally incongruent elders, generic interiors, or distorted facial details, including several instances where Sora rendered the grandfather as a Western man in

a Western-style kitchen, producing discomfort and rendering the videos unsafe to share with family. Yet small, incidental micro-cues such as a brief smile at the start of one clip proved unexpectedly resonant. Technically, the trial showed that complex narrative prompts overloaded current models, flattening emotional nuance and destabilizing identity representation. The switch from Sora to Doubao further highlighted how cultural fit in training data materially shapes emotional plausibility. Conceptually, this trial revealed that autobiographical memory does not function as a coherent scene that can be reconstructed. Instead, remembrance is anchored in small, meaningful traces that surface through attempted reconstruction.

These insights made clear that full-scene reconstruction exceeded what current tools could meaningfully support, both technically and emotionally. The contrast between failed narratives and resonant micro-expressions motivated a deliberate narrowing of scope. The next trial therefore shifted toward a minimal memorial form focused on a single facial expression, an exploratory move to test whether small, controllable cues could better sustain emotional fidelity than complex scenarios.

4.1.2 Expression Animation as a Minimal Memorial Form. Building on the realization that micro-moments carried more emotional weight than full scenarios, this trial focused on animating a subtle smile from a single portrait. Using Doubao, the researcher generated short clips in which the grandfather briefly shifted from neutrality into a gentle smile. Across attempts, the prompts were gradually refined to constrain exaggeration and ensure the smile faded quickly, matching how he typically expressed emotion.

Across these attempts, the emotional fit improved markedly. The micro-accuracy of the slight smile evoked stronger resonance than any full-scene reconstruction in Trial 1, and family dialogue affirmed this alignment. The mother confirmed the smile felt similar but noted that he would always return to a serious expression quickly, which informed further refinements. Technically, the trial showed that AI systems handle expression-level animation far more reliably than complex scenarios, and that even small exaggerations must be managed explicitly in prompts. Conceptually, the trial showed that DIY memorialization can build on these traces by animating subtle expressions, reframing AI not as a scene simulator but as a reflective tool that amplifies emotionally salient cues.

Despite the technical success around our intention, the model repeatedly erased the grandfather's missing arm, unintentionally removing a defining part of his embodied identity. This highlighted that facial cues alone were insufficient for representing the fullness of memory and that bodily characteristics such as posture, gait, sleeve position carry equally meaningful traces. These insights motivated a shift toward exploring movement and embodied cues in the next trial, prompting Trial 3's focus on embodied rendering.

4.1.3 Movement Rendering as an Embodied Memorial Form. Building on the insight that subtle cues hold emotional power, this trial explored whether embodied traits, particularly movement and embodied cues, could be rendered meaningfully. Using Doubao, the researcher generated short clips of the grandfather walking, specifying a thin, upright figure with an empty right sleeve, and stable pacing. Multiple attempts ranged from back-view reconstructions inferred from a frontal photo to grandparents walking scenes inspired by an old family photograph.

Across attempts, small distortions in gait, such as overly light, playful motion, felt emotionally violating, contradicting memories of the grandfather's seriousness and dignity. Technically, the trial revealed the instability of asking models to infer unseen views or fabricate scenarios not anchored directly in photographic material. Yet the two-person walking clip introduced a different form of resonance: although the two-person walking clip inaccurately portrayed the grandparents as closer and more coordinated than they were in life, the shared rhythm of their movement produced a gentle, wishful version of their relationship that still felt emotionally acceptable to both the researcher and her mother. Conceptually, the trial showed that embodied cues do more than preserve memory, but they open space for reinterpretation, revealing how movement can both anchor and subtly reshape remembrance.

A turning point came when the mother noted that the grandfather had both arms in his youth, reframing the missing arm not as a fixed identity marker but as a life-stage-specific reality. This, combined with the insight that movement can support emotionally desirable reinterpretations, motivated a shift toward speculative reconstruction, imagining the grandparents in their youth rather than reproducing familiar scenes. This led to Trial 4's exploration of youthful reinterpretation as an alternative memorial form.

4.1.4 Youth Reinterpretation as a Speculative Memorial Form. This final trial explored a speculative direction that imagined the grandparents in their youth. Rather than attempting to recover factual appearance, the researcher framed the task as an artistic reconstruction inspired by an elderly portrait. Using Doubao, a series of iterations adjusted age regression, clothing from 1950s-60s China, hairstyle, posture, and subtle expressions. Constraints were added to ensure respectful distance between the figures, gentle movements, and the presence of two healthy arms, reflecting how the grandfather would have appeared before his accident in his 20th.

The trial revealed a different mode of emotional resonance: unlike earlier iterations, the connection emerged not from accuracy but from plausibility. The youthful reinterpretations offered a gentle possible past that felt temperamentally true even without grounding in lived memory. Iterations highlighted a few model sensitivities: age regression required strong constraints, hairstyles anchored temporal authenticity, and the reference photo biased the system toward the later-life disability. Once corrected, the appearance of two healthy arms generated relief rather than disruption, showing how speculative reconstruction can restore aspects of identity that memory alone cannot. Conceptually, this trial shifted DIY memorialization from reanimating memories to imagining unrealized temporalities, suggesting that remembrance may also draw strength from aspirational or restorative visions.

Youth reinterpretation marked a natural culmination of the design trajectory. Whereas the earlier trials sought to refine how AI might faithfully reanimate remembered cues, this trial demonstrated that AI can also support meaning-making by enabling families to imagine temporally distant, even restorative, versions of loved ones. Rather than opening a new direction for exploration, this speculative shift reframed the purpose of the entire inquiry, showing that AI-mediated DIY memorialization extends beyond reconstructing what was remembered to creatively refiguring what

could have been. Combined with the insights across trials, the trajectory reached a provisional sense of completion here.

4.2 Cross-Iterative Insights

Across the four trials, the design explorations collectively revealed how AI-mediated DIY memorialization generates insights across emotional, relational, technical, and conceptual dimensions.

4.2.1 Emotional-Realistic-Ethical Boundaries. Across trials, emotional fit consistently hinged on micro-level cues, such as subtle gestures, a brief smile, and a familiar walking rhythm. Misrepresentations, especially in culturally salient features or bodily identity, produced immediate emotional and ethical breaches. Yet some “inaccuracies” became emotionally restorative reinterpretations, such as restoring the grandfather’s arm or softening relational distance, showing that acceptable inaccuracy is negotiated, not predetermined. These boundaries illustrate that AI-mediated DIY memorialization relies on emotional truth more than factual precision.

4.2.2 Family Co-Memory and Interpretation Construction. Selective sharing emerged as a form of curation, determining which AI outputs entered the shared grieving space and which remained private. Once shared, the videos acted as conversational objects that invited memory correction, helped family members fill gaps in autobiographical memory, and supported reinterpretations. The creation process itself became a trigger for multigenerational remembrance, revealing how family voices stabilize, contest, and expand the meaning of AI-generated media. This highlights that AI-mediated memorialization is inherently relational and co-authored.

4.2.3 AI as Design Material. Generative models are reliable when animating what already exists, such as micro-expressions and slight posture shifts, but fragile when asked to infer unseen information, such as complex actions from a frontal portrait. Overly complex prompts consistently overloaded models, confirming the need for minimal viable prompts rather than narrative reconstruction. Cultural mismatches, age-regression failures, and unstable limb rendering exposed the limits of treating AI as a factual simulator. Instead, AI functioned as a fragile yet generative design material whose breakdowns clarified which memory elements were emotionally non-negotiable and which could be reinterpreted.

4.2.4 Conceptual Shifts in AI-Mediated DIY Memorialization. Across trials, AI-mediated DIY memorialization shifted from attempting to reconstruct remembered episodes toward selectively authoring emotionally meaningful and at times speculative forms. The researcher’s ongoing acceptance, rejection, and fine-tuning of outputs illustrated a mode of authorship grounded in emotional fit rather than factual accuracy. In practice, AI functioned less as a historical simulator and more as an unreliable yet generative collaborator that surfaced salient memory cues and opened new representational possibilities, including “possible pasts” marked by youthful vitality, relational warmth, or restored identity. Collectively, these insights frame AI-mediated DIY memorialization as a hybrid practice of remembering and refiguring, where AI-generated media become materials for maintaining continuing bonds, rooted in lived memory yet open to gentle, restorative reinterpretation.

4.3 Design Implications

AI-mediated DIY memorialization requires designing not for photo-realistic fidelity, but for emotional and ethical safety, stability, and relational meaning-making. Four implications have emerged.

4.3.1 Prioritize minimal, high-fidelity cues. Emotional resonance consistently hinged on micro-level signals, such as subtle expressions, posture, pacing, rather than full-scene accuracy. Complex reconstructions overloaded generative models and introduced misaligned details that caused ethical breaches. Systems should therefore adopt a minimal-first generation strategy, foregrounding micro-cues as primary memorial materials before expanding to more complex compositions. Supporting users in intentionally layering detail, rather than producing entire scenes at once, aligns with how remembrance itself operates through selective traces rather than narrative completeness.

4.3.2 Support user-defined memory boundaries. Trials showed that some elements of identity (e.g., facial expression, gait) are emotionally non-negotiable, while others (e.g., relational closeness, youthful vitality) may be open to restorative reinterpretation. Systems should therefore help users articulate personal memory boundaries and differentiate between: (1) aspects that must be preserved for emotional safety, and (2) aspects where reinterpretation may be welcomed or generative. Tools can communicate where AI is stable versus fragile, and allow boundary-setting to dynamically evolve as users reflect. Designing for both protection and intentional reinterpretation acknowledges that memorial meaning is not fixed but negotiated.

4.3.3 Scaffold multi-voice reflection. Memorial meaning often emerges relationally, as different family members bring distinct memories, corrections, and reinterpretations. Systems should therefore treat AI-generated media as prompts for shared reflection rather than fixed outputs. This can include features that support annotating or responding to clips, and inviting multiple perspectives without forcing consensus. By framing outputs as conversational objects, systems can help families co-construct meaning while preserving emotional safety and individual boundaries.

4.3.4 Attend to cultural fit in model behavior. Generative models encode implicit cultural assumptions that influence facial appearance, interior spaces, clothing, and bodily style, factors that strongly shape emotional plausibility in memorial contexts. Systems should therefore surface cultural biases in model behavior and support users in choosing or tuning models that align with their lived contexts. This may involve offering culturally adaptive presets, warning users when outputs deviate from expected cultural cues, or allowing users to anchor generation more firmly in culturally specific reference materials. Designing for cultural fit helps ensure that outputs feel respectful, relatable, and emotionally safe.

4.4 Reflections on RtD in Sensitive Contexts

Conducting RtD in domains of loss and remembrance revealed methodological dynamics distinct from other design settings. First, *design breakdowns were not neutral technical failures but emotionally charged events that exposed ethical boundaries and memory sensitivities*. Misaligned facial features, playful gait, or culturally

incongruent appearances produced immediate discomfort, demonstrating that evaluation is inseparable from emotional impact. Such moments became analytic turning points, redirecting the design trajectory more forcefully than planned criteria ever could.

Second, *the designer's subjectivity, including memories, hesitations, relief, and moments of emotional breach, functioned not as noise but as core empirical material.* The designer's reactions provided the earliest detection of unsafe outputs and guided the fine-grained identification of which cues mattered. This aligns with autobiographical design commitments that lived experience is the primary site of insight, especially concerning sensitive contexts.

Third, *RtD progress proved non-linear and required active safeguarding of emotional well-being.* Some iterations demanded slowing down, pausing, or entirely abandoning a direction when it crossed emotional thresholds. The "stop moments" were not methodological weaknesses. Instead, they clarified the limits of what should be designed or reanimated. In this way, emotional safety became a design constraint and an analytic lens in the RtD process.

Finally, *ethical permissibility emerged dynamically rather than at the outset.* What felt acceptable or restorative, such as restoring the grandfather's arm or imagining a youthful past, could not have been predicted beforehand. The research process itself surfaced which reinterpretations supported remembrance and which undermined it. This underscores that in ethically charged RtD, ethical understanding evolves through making, breakdown, reinterpretation, and dialogue rather than through static principles.

In summary, these reflections show that RtD in memorial contexts requires heightened attunement to emotional signals, tolerance for non-linearity, and a methodological stance that treats subjectivity and vulnerability not as limitations but as essential pathways to knowledge.

5 Anticipated Contribution

This study is expected to contribute to the HCI and design community in three interconnected ways. First, this study develops the emerging concept of AI-mediated DIY memorialization through a situated, iterative exploration of how individuals actually work with generative AI to remember, reinterpret, and refigure the deceased. Across four trials, the inquiry shifted from attempting narrative reconstruction toward selectively authoring emotionally meaningful micro-cues and, eventually, speculative youth reinterpretations. This evolution provides a grounded account of how memorial meaning is constructed through emotional-fit authorship, iterative refinement, and gentle reinterpretation. Thus, it deepens conceptual understanding of how AI-generated videos can support remembrance and continuing bonds.

Second, this study contributes methodological insight into conducting Research through Design in sensitive contexts. The work demonstrates how design breakdowns, emotional breaches, and moments of hesitation function as analytic material that redirects inquiry more strongly than predefined criteria. It illustrates how the designer-researcher's subjectivity can serve as empirical data for reasoning about ethically complex design spaces. These reflections extend autobiographical design methodology by foregrounding emotional safety as both a design constraint and an analytic lens,

emphasizing non-linearity, and showing how ethical permissibility evolves dynamically through making and reflection.

Third, this study produces actionable design implications for future memorial technologies. Through iterative engagement with generative models, the work identifies the value of privileging minimal, high-fidelity cues; supporting user-defined memory boundaries; enabling multi-voice reflection; and attending to cultural fit in model behavior. These implications offer practical guidance for researchers and designers building systems that support DIY memorial creation, prioritizing emotional grounding and relational meaning-making over photorealistic accuracy and completeness.

In summary, this study positions AI-mediated DIY memorialization as a promising and ethically sensitive design space. Through its conceptual, methodological, and practical contributions, this work advances understanding of how generative AI may support remembrance and continuing bonds in ways that honor emotional nuance, relational dynamics, and cultural specificity.

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This project involved the use of generative AI systems (including Sora and Doubao) as part of the research and design process. These tools were used to generate AI-produced video artifacts that served as design probes for exploring AI-mediated DIY memorialization. All outputs were critically examined through iterative reflection, and no AI-generated content was used without human evaluation, interpretation, and intervention.

Portions of the writing were supported by generative AI tools (specifically OpenAI's ChatGPT) for language refinement, grammar checking, and clarity improvements. All conceptual insights, analytical framing, methodological decisions, and final arguments are my own. AI assistance was limited to surface-level expression and did not contribute original ideas or interpretations.

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